

MID-TERM EXAMINATION

Course: Applied Statistics for Engineers and Scientists	
Time: 60 minutes	Term: 2 – Academic year: 2021-2022
Lecturer(s): Dr. Le Thi Xuan Mai	
Student name:	Student ID:

(Notes: All of paper document are permitted)

Exercise 1: (3 marks) A package of 6 fuses are tested with the probability an individual fuse is defective is 0.05 that is 5% of all fuses manufactured are defective.

- a. What is the probability at least one fuse will be defective?
- b. What is the probability that more than one fuse will be defective, given that at least one is defective?

Exercise 2: (2 marks) A bag I contains 4 white and 6 black balls while another bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from Bag I.

Exercise 3: (3 marks) Preventing fatigue crack propagation in aircraft structures is an important element of aircraft safety. An engineering study to investigate fatigue crack in $n=13$ cyclically loaded wing boxes reported the following crack lengths (in mm):

Crack length (in mm) x_i	Frequency N_i
1.15	1
1.37	3
1.83	2
2.04	1
2.13	1
2.47	1
2.60	1
2.96	1
3.02	2
Total	13

- a. Calculate the arithmetic mean, mode.
- b. Calculate the frequency and relative frequency.
- c. Calculate the median, first quartile, third quartile and interquartile range.

Exercise 4: (2 marks) Suppose a distribution has mean $\mu=19.2$ and standard deviation $\sigma=3.2$. What the Chebyshev's Theorem tell us about the probability a randomly selected value is between 12.8 and 25.6?

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